

Laplace (continued)

By

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Initial and Final Value Theorems

- Initial value theorem

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s),$$

- Final value theorem

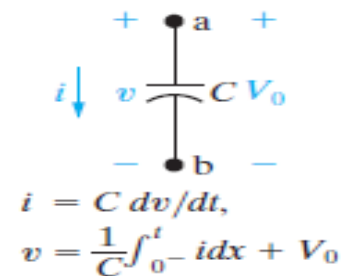
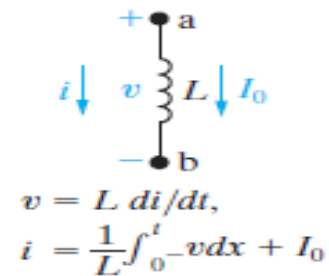
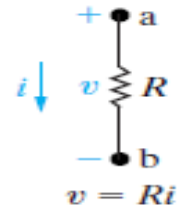
$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s).$$

Different partial fraction techniques(summery)

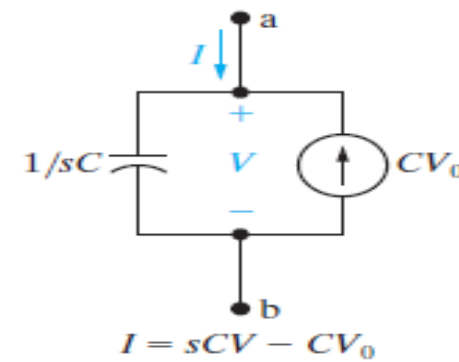
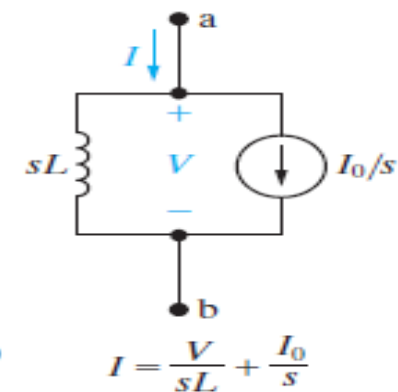
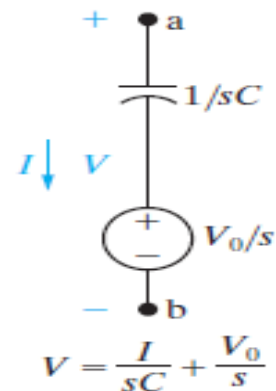
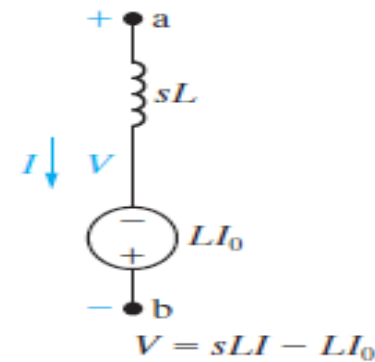
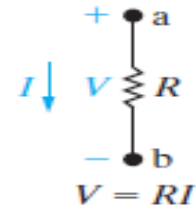
Pair Number	Nature of Roots	$F(s)$	$f(t)$
1	Distinct real	$\frac{K}{s + a}$	$Ke^{-at}u(t)$
2	Repeated real	$\frac{K}{(s + a)^2}$	$Kte^{-at}u(t)$
3	Distinct complex	$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$	$2 K e^{-\alpha t} \cos(\beta t + \theta)u(t)$
4	Repeated complex	$\frac{K}{(s + \alpha - j\beta)^2} + \frac{K^*}{(s + \alpha + j\beta)^2}$	$2t K e^{-\alpha t} \cos(\beta t + \theta)u(t)$

Circuit elements in s domain

TIME DOMAIN

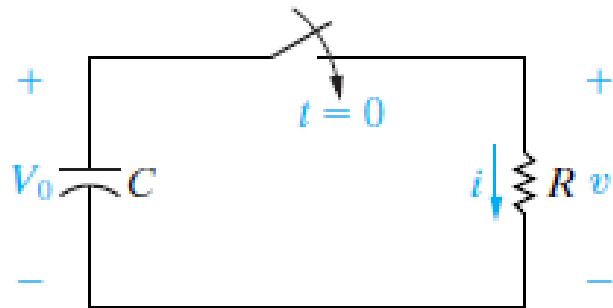


FREQUENCY DOMAIN

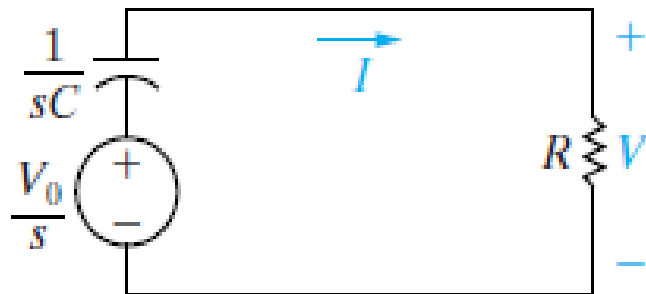


Natural response of an RC circuit using Laplace

- Capacitor is initially charged to V_0 , and switch is closed



- S domain equivalent circuit



- Writing equation on the loop and taking inverse transform

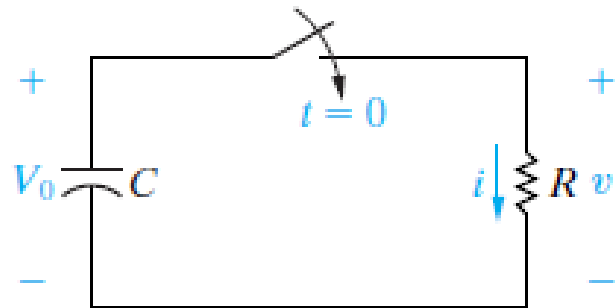
$$\frac{V_0}{s} = \frac{1}{sC}I + RI.$$
$$I = \frac{CV_0}{RCs + 1} = \frac{V_0/R}{s + (1/RC)}.$$

$$i = \frac{V_0}{R}e^{-t/RC}u(t),$$

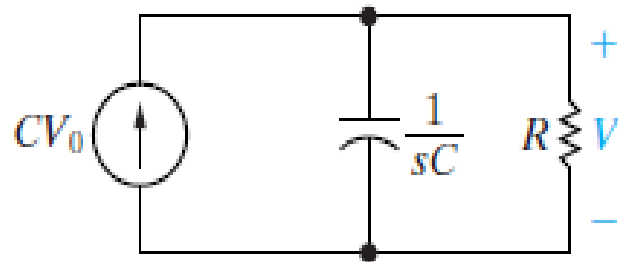
$$v = Ri = V_0e^{-t/RC}u(t).$$

Natural response of an RC circuit using Laplace(2nd method)

- Capacitor is initially charged to V_0 , and switch is closed



- S domain equivalent circuit



- Writing equation on the node and taking inverse transform

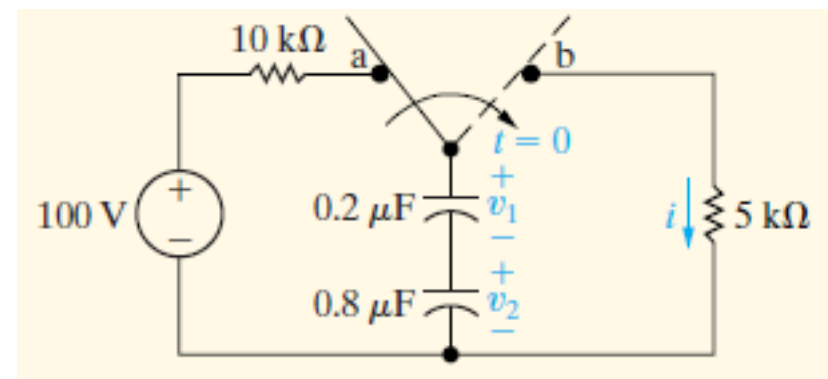
$$\frac{V}{R} + sCV = CV_0.$$

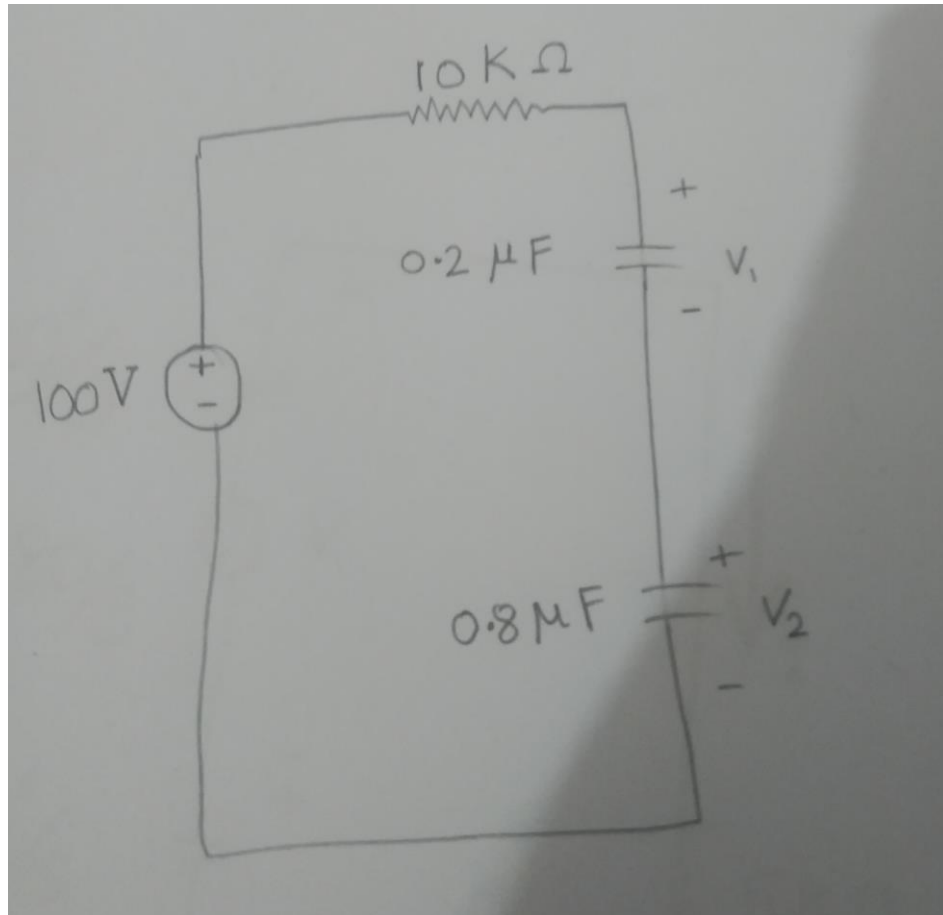
$$V = \frac{V_0}{s + (1/RC)}.$$

$$v = V_0 e^{-t/RC} = V_0 e^{-t/\tau} u(t).$$

The switch in the circuit shown has been in position a for a long time. At $t = 0$, the switch is thrown to position b.

- Find I , V_1 , and V_2 as rational functions of s .
- Find the time-domain expressions for i , v_1 , and v_2 .





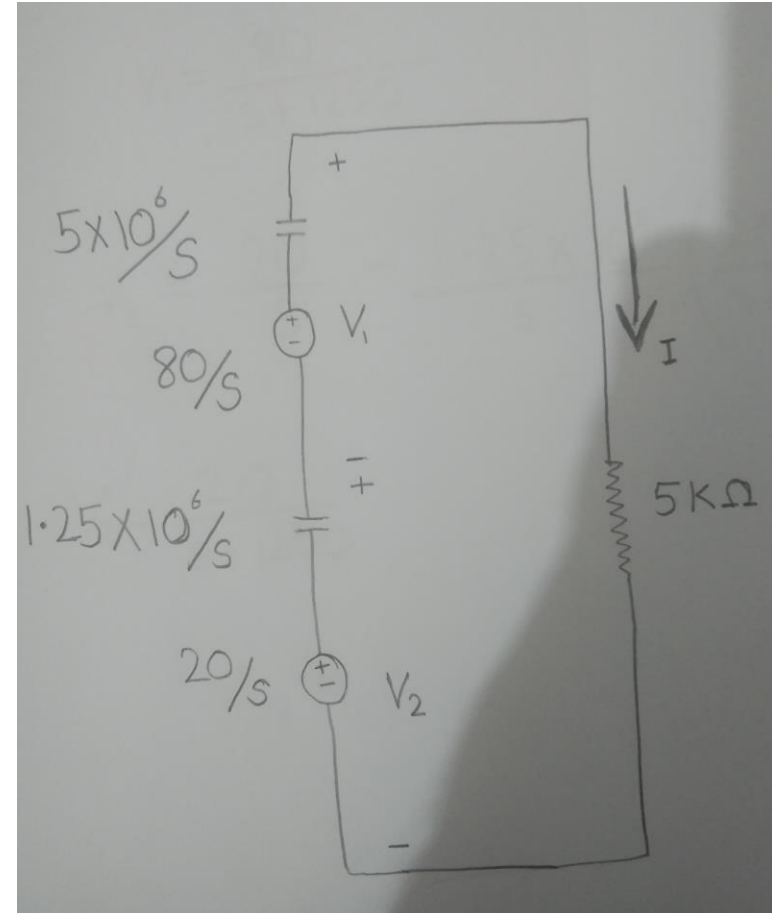
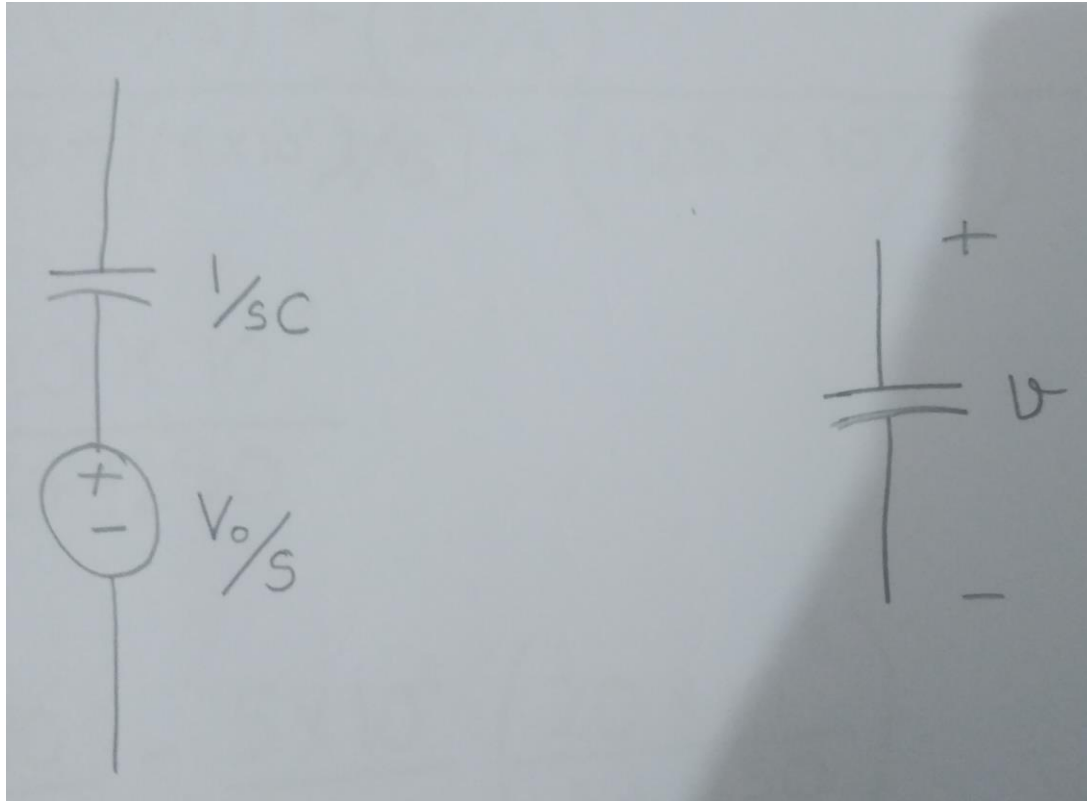
At $t = 0^-$, $0.2 V_1 = (0.8) V_2$;

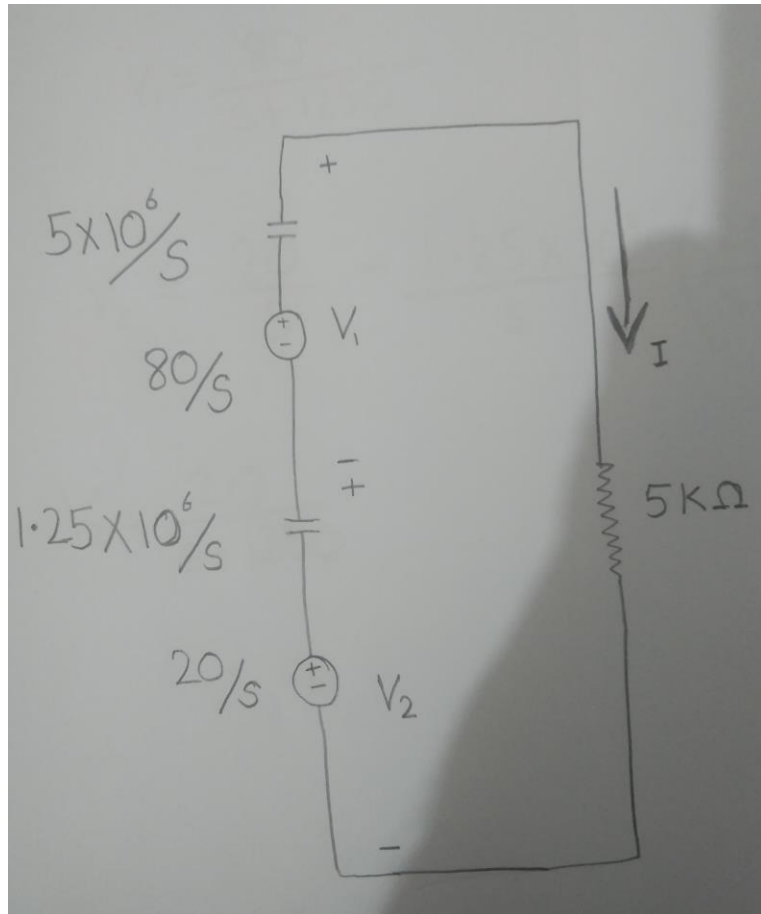
$$V_1 = 4 V_2; \quad V_1 + V_2 = 100 \text{ V}$$

Therefore,

$$V_1(0^-) = 80 \text{ V} = V_1(0^+);$$

$$V_2(0^-) = 20 \text{ V} = V_2(0^+)$$





$$I = \frac{(80/s) + (20/s)}{5000 + [(5 \times 10^6)/s] + (1.25 \times 10^6/s)}$$

$$I = \frac{20 \times 10^{-3}}{s + 1250}$$

$$V_1 = \frac{80}{s} - \frac{5 \times 10^6}{s} \left(\frac{20 \times 10^{-3}}{s + 1250} \right)$$

$$V_1 = \frac{80}{s + 1250}$$

$$V_1 = \frac{80}{s+1250}$$

$$V_2 = \frac{20}{s} - \frac{1.25 \times 10^6}{s} \left(\frac{20 \times 10^{-3}}{s+1250} \right)$$

$$V_2 = \frac{20}{s+1250}$$

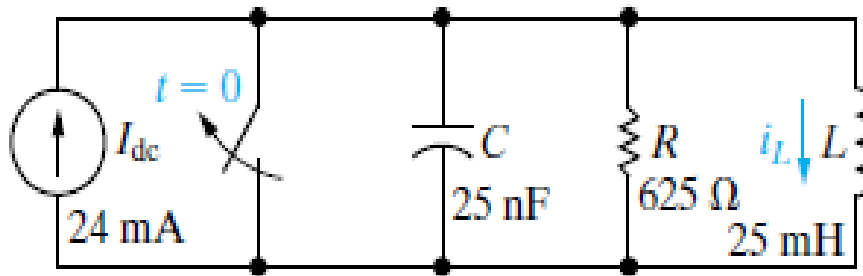
$$i = 20 e^{-1250t} u(t) \text{ mA};$$

$$V_1 = 80 e^{-1250t} u(t) \text{ V}$$

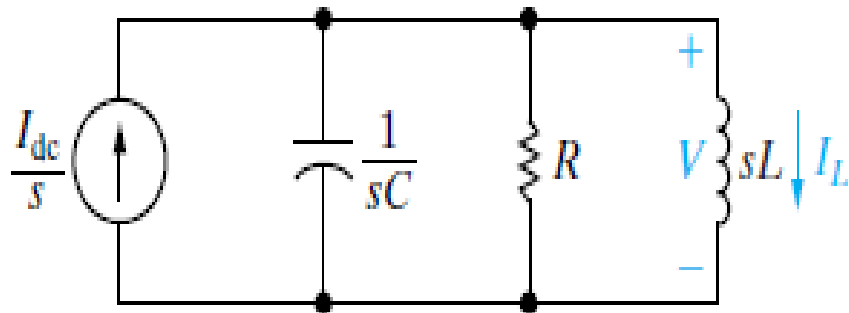
$$V_2 = 20 e^{-1250t} u(t) \text{ V}$$

Step response of parallel circuit

- Time domain circuit



- S domain circuit is below



- Inductor current and node eq

$$I_L = \frac{V}{sL}$$

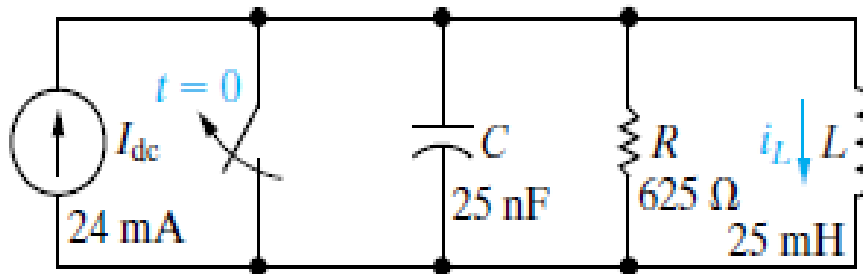
$$sCV + \frac{V}{R} + \frac{V}{sL} = \frac{I_{dc}}{s}$$

$$V = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$$

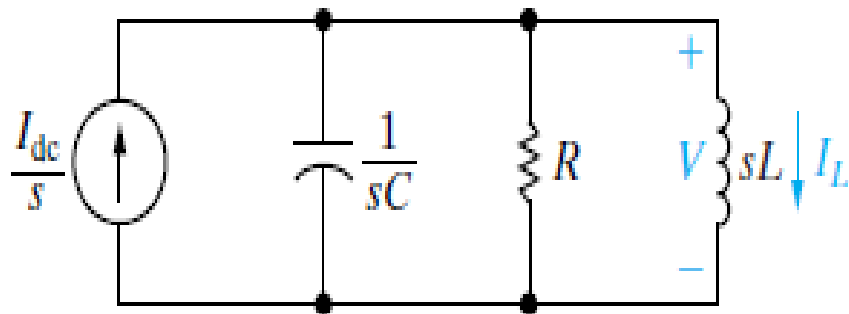
$$I_L = \frac{I_{dc}/LC}{s[s^2 + (1/RC)s + (1/LC)]}$$

Step response of parallel circuit(actual values)

- Time domain circuit



- S domain circuit is below



- Using actual values

$$I_L = \frac{I_{dc}/LC}{s[s^2 + (1/RC)s + (1/LC)]}$$

$$I_L = \frac{384 \times 10^5}{s(s^2 + 64,000s + 16 \times 10^8)}$$

$$I_L = \frac{384 \times 10^5}{s(s + 32,000 - j24,000)(s + 32,000 + j24,000)}$$

Step response of parallel circuit(actual values)

- Using final value theorem

$$\lim_{s \rightarrow 0} sI_L = \frac{384 \times 10^5}{16 \times 10^8} = 24 \text{ mA.}$$

$$I_L = \frac{K_1}{s} + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}.$$

$$K_1 = \frac{384 \times 10^5}{16 \times 10^8} = 24 \times 10^{-3},$$

$$K_2 = \frac{384 \times 10^5}{(-32,000 + j24,000)(j48,000)} = 20 \times 10^{-3} \angle 126.87^\circ.$$

$$i_L = [24 + 40e^{-32,000t} \cos(24,000t + 126.87^\circ)]u(t) \text{ mA.}$$